

Social exchange in buyer-supplier relationships and innovation speed: the mediating and moderating role of information sharing and knowledge channels

Divesh Ojha, Mumin Dayan, Beth Struckell, Amandeep Dhir and Terrence Pohlen

Abstract

Purpose – This study recognizes service as the majority contributor to global and US gross domestic product and the importance of innovation speed to service innovation. Generating innovative products and services at a faster rate generates advantages for business-to-business (B2B) service organizations in keeping up with and moving ahead of rivals. This study aims to introduce the concept of capacity for social exchange (CSE) in buyer–supplier relationships, which reflects the degree to which individuals possess competencies that enable the exchange of information, and this study also explores how CSE affects knowledge sharing and innovation speed within a supply chain organization.

Design/methodology/approach – The sampling frame of this research consisted of service businesses in the USA. The data were collected through Zoomerang, an online survey research firm where the B2B panel of Zoomerang formed the sampling frame. The data was collected from 264 B2B service sector executives.

Findings – The key findings are as follows: CSE facilitates knowledge sharing; knowledge sharing is positively related to innovation speed; and the relationship between CSE and innovation speed is fully mediated by knowledge sharing. Additional analysis reveals that knowledge channels which are designed as boundary-spanning strategies used to transfer and create knowledge between business units, directly increase – and positively moderate – the relationship between CSE and innovation speed.

Originality/value – This study addresses the gap in literature focused on micro-level influencers on innovation. This study sets out by presenting the concept of CSE, and this study addresses limitations in prior work by examining the research questions.

Keywords Capacity for social exchange, Knowledge sharing, Knowledge channels, Innovation speed

Paper type Research paper

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1. Introduction

Service firms, which currently represent more than 70% of global gross domestic product (Chen *et al.*, 2016), operate in an increasingly dynamic environment that subjects them to intense pressure to be highly responsive to competitive service offerings and continuously launch novel offerings (Berry *et al.*, 2006). Business-to-business (B2B) service businesses face unique challenges in delivering innovation in a landscape inhabited by traditional product manufacturers. Easily copied by competitors, B2B service innovation demands continuous and rapid bursts of innovation activity (Kessler and Chakrabarti, 1996). Competitive advantage in this environment is therefore crucially dependent on a company's embedded innovation speed (Kessler and Chakrabarti, 1996; Chen *et al.*, 2012); however, faster service innovation is not always a straightforward process for B2B firms (Thomas, 2013). This has led to service innovation becoming a process that is interconnected with

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outside entities (Ndubisi *et al.*, 2020; Markovic *et al.*, 2020). Prior research has revealed that a constant flow of information is required to ensure faster innovation when incorporating suppliers into the innovation process (Zhao *et al.*, 2020; Song and Di Benedetto, 2008). Consequently, as supplier involvement becomes more significant in the process, the role of B2B firms' knowledge exchange capabilities in developing faster innovation becomes an important research topic (Alinaghian *et al.*, 2020; Yang *et al.*, 2019).

Given the importance of innovation speed, researchers have explored multifarious mechanisms to influence innovation speed within organizations. Although several studies have examined antecedents to innovation speed, much of the interest has been centered on macro-level variables. Kessler and Chakrabarti (1996), for example, explore how a firm's strategic orientation, goal alignment and organizational capabilities affect innovation speed. Chen *et al.* (2010) and Struckell and Ojha (2017) focus on the role of leadership, bricolage and operational flexibility in enabling innovation. However, there is a dearth of studies on how individual-level or micro-level phenomena affect innovation speed. Further, those studies focus on innovation and knowledge sharing which do include individual-level variables and neglect broader, more generic conceptualizations to focus instead on specific concepts such as self-efficacy, which can hamper theory-building efforts. Despite revealing the importance of theorizing the actions and interactions of individuals in buyer–supplier relationships (Liu *et al.*, 2017; Forkmann *et al.*, 2018), and given the wide recognition of the role played by individual capabilities, such as relationship termination capability in B2B relationships and networks (Zaefarian *et al.*, 2017), it is striking that recent research has yet to fully examine this phenomenon. According to Foss and Lindenberg (2013), micro-foundations involve theorizing in terms of the actions and interactions of individuals. Lack of emphasis at this level is concerning, as it is these variables that have the greatest potential to generate novel conclusions and suggest informed interventions in buyer–supplier relationships. For instance, Alinaghian *et al.*'s (2020) case study reveals that joint practices in a B2B network are noticeable only when both the socially and cognitively embedded conditions accurately become established in a given relationship. Specifically, without joint optimistic attitudes and motivation at the micro-level, buyer–supplier relationships can hardly be beneficial to both parties.

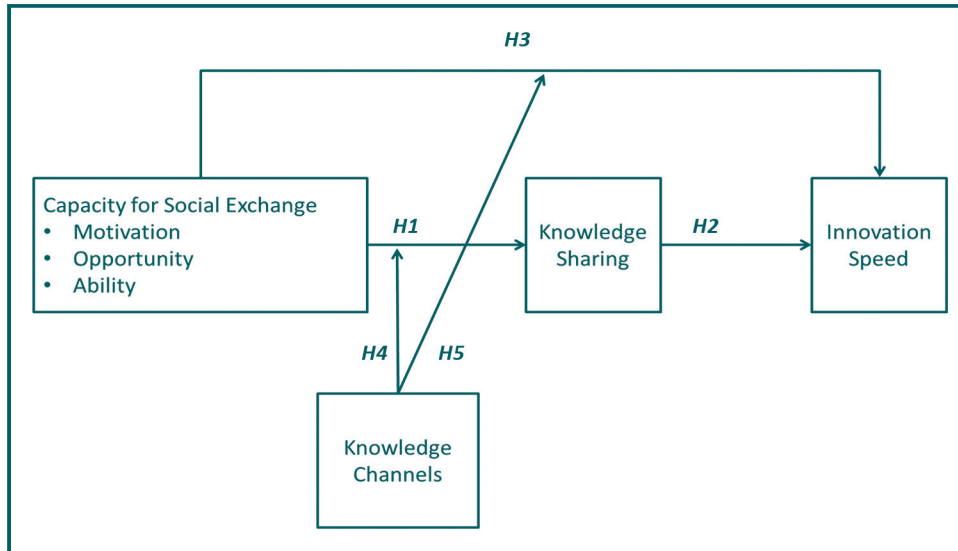
To address these limitations as well answer numerous calls for studies on these issues (Forkmann *et al.*, 2018; Alghababsheh and Galleary, 2020), we introduce the notion of “Capacity for Social Exchange” (CSE) in buyer–supplier relationships, which we define as the degree to which individuals possess a set of competencies that allow them to share and exchange knowledge with others. By using the lens of social exchange theory, we examine how CSE affects the sharing of knowledge and interacts with the organizational context so as to influence speed of innovation. Social exchange theory has been recognized as a promising perspective within the knowledge management literature and is uniquely useful because it provides holistic arguments for when such exchanges are likely to occur (Wu *et al.*, 2014).

This contribution builds upon our conceptualization of CSE. We set out by presenting the concept of CSE, and we address limitations in prior work by examining our research questions. We first question how CSE affects knowledge sharing in our B2B service innovation context (i.e. between business units of an organization and its supply chain partners). Second, we ask how knowledge sharing affects innovation speed and whether, as expected, sharing entirely mediates the effect of CSE on innovation speed. Finally, by using structural equation modeling (SEM) to test our hypotheses, we explore how the use of integrating mechanisms (i.e. knowledge channels [KC] that allow for exchange and creation of knowledge) enhances knowledge sharing and innovation speed in organizations. We conclude by discussing our study's contributions as well as its limitations.

2. Theoretical framework

Figure 1 presents our theoretical model, and we proceed to discuss the various elements contained therein.

Figure 1 Theoretical model for social exchange and innovation speed



2.1 Capacity for social exchange

While the development of knowledge as a resource within organizations is undeniably influenced by a broad range of factors, organization theorists have lamented the overt emphasis that has been placed on studying more macro and higher-level phenomena, such as competitive intensity, environmental turbulence, absorptive capacity and strategic orientation, among others (Foss *et al.*, 2010). The dearth of studies that focus on micro-foundations (i.e. individual-level phenomena and their interaction within organizational contexts) is concerning because, ontologically, the development of knowledge as a resource is contingent on individual attitudes, behaviors and actions. Prior work suggests that neglecting micro-foundations poses a potentially serious deficit for the knowledge management literature, to the extent that “understanding what happens in organizations requires beginning from their fundamental constituents, namely individuals” (Foss *et al.*, 2010, p. 457), as well as generating insight at the level of individuals (i.e. organizational members) and their mutual interaction within their context. Emphasizing micro-levels means identifying the true mechanisms that produce observed correlations, and an understanding of such generative mechanisms generates the basis for informed intervention (Felin and Foss, 2005; Brenton and Levin, 2012).

It follows that the limitations in the available knowledge-based literature affect the definition of the level and domain of the concept at hand, and therefore we define the CSE as a broad set of competencies which enable individuals to interact with others and share or exchange knowledge. The level of the CSE concept as an individual micro-foundation remains important because, ultimately, all organizational action and intervention occurs at the level of the individual. The individual level domain of CSE is crucial, for the sharing of knowledge, which has ultimately been associated with improved organizational performance (Della Peruta *et al.*, 2014; Chang, 2017; Yang *et al.*, 2019; Rungsithong and Meyer, 2020). Sharing knowledge (at the individual level) can lead to increased absorptive capacity and the development of other organization level capabilities. Previous studies have proposed that understanding how knowledge is shared is important, as it is through interaction between organizational members that an overlap of knowledge sets occurs and absorptive capacity is increased, which, in turn, improves the identification, reconciliation and prompt responses to problems (Forkmann *et al.*, 2018; Kohtamäki *et al.*, 2018; Argote and Miron-Spektor, 2011).

We adopt the perspective that knowledge is embedded in individuals, and hence that organizational performance can only be fully understood by examining action at this level (Wasko and Faraj, 2005). A number of studies have focused on individual-level determinants of knowledge sharing and performance. For example, Hendriks (1999) posits that sharing is motivated by the desire to be successful, whereas Eze *et al.* (2013) examine how reward systems, confidence and leadership qualities encourage individuals to share knowledge with others. Although a wide range of theoretical perspectives and individual constructs have provided valuable research contributions, for reasons of practicality and necessity, these have focused on specific, individual-level variables based upon the specific theoretical paradigm used and their relationship to knowledge sharing and performance. A few studies have emphasized the role of extrinsic or intrinsic rewards in defining attitudes toward knowledge sharing; for instance, Jeon *et al.* (2011) use the theory of planned behavior to explore how both “enthusiasm to help” and “perceptions of reciprocity,” motivate individuals and affect knowledge sharing. Although it is important to understand how enjoyment and reciprocity facilitate sharing, it becomes increasingly clear that these factors alone do not suffice to share knowledge (Blau, 1964; Serenko and Bontis, 2016). Additionally, individuals need to believe that the knowledge they hold is valuable, accurate and acceptable within the organization. Without confidence in the knowledge they hold, motivation may not suffice and, thus, may limit the will to share and diminish the perceived benefits in helping organizations improve performance.

A further issue pertaining to knowledge management studies at the level of the individual, derives from the diversity in the use and conceptualization of constructs, as well as the inconsistency in study findings. While Kim and Lee (2006) found that reward systems significantly impact knowledge-sharing capabilities, Lin (2007) reported that organizational rewards did not have an effect on employees' willingness to share knowledge with their colleagues. Such contradictory findings are problematic, as they limit practitioner recommendations and impede theoretical interpretation, synthesis and theory development for research.

Considering these observations, yet while recognizing the value of prior research, our objective here is to enrich the knowledge management literature and aid in further theory development by introducing the broad, higher-order concept of individual CSE. CSE reflects the set of individual competencies necessary to engage and interact with others, with the intent to share knowledge. Specifically, we draw upon the framework introduced by MacInnis and Jaworski (1989) in the context of marketing and use the three dimensions of motivation, opportunity and ability (MOA) to reflect the dimensions of CSE. These three dimensions have been insinuated in numerous studies, as well as in theoretical perspectives, that explore the role of individual-level factors in knowledge sharing and organizational innovation. Here, these dimensions are appropriate for two reasons. First, they allow us to step back and adopt a broader and more abstract perspective to avoid the contradictions and problems of inconsistency that often surface when theorizing from more specific variables (such as rewards). As Foss and Lindenberg (2013) contend, and because all phenomena are mediated through micro-foundations (e.g. organizational culture > affects motivation > which affects performance), putting the spotlight on micro-foundations of interest can alleviate difficulties related to the use of the divergent concepts and inexplicable contradictions that have appeared in similar studies. Second, the motivation, opportunity and ability dimensions allow us to propose a simple yet holistic concept that includes those variables necessary to facilitate knowledge sharing – as suggested in prior research.

2.2 Composition of capacity for social exchange: theoretical perspectives and empirical studies

2.2.1 Social exchange theory. The knowledge management literature predominantly adopts the perspective of social exchange theory, where individual behavior and individual

interactions with others are based upon a self-interest analysis of the costs and benefits of the interaction (Lambe *et al.*, 2001). Individuals seek to maximize benefits and minimize costs when interacting with others. While initially framed as a strictly economic exchange, later developments in social exchange theory have illuminated that benefits and costs can be intangible and may be realized in either a short- or long-term manner. The theory suggests that individuals build relationships and exchange knowledge with others to maximize benefits with expectations of future returns. For example, Davenport and Prusak (1998) suggest reciprocity, promotions and status as potential intangible and longer-term motivators for behavior. In a more recent B2B study, Cortez and Johnston (2020) explain social exchange theory to encompass economic and or social outcomes which can manifest in both near-term and longer-term outcomes.

On the other hand, Knowledge sharing and exchange have been found to have costs along with the associated benefits. The cost represents the risk associated with knowledge exchange. Individuals may be discouraged if they perceive risk in interacting with others. Organizational cultures that emphasize outcomes may discourage sharing because individuals perceive that knowledge-sharing interactions may result in the loss of a personal competitive edge, the possibility of being made redundant and losing work or the risk management interpretation of individuals having “excess time on the organizational dime.” If these issues lead management to perceive an individual in a negative way, the consequences are viewed as significant: loss of work or benefits.

Prior literature covers some aspects of the importance of individual motivations to share knowledge. Roodbari *et al.* (2016a) provide a small yet representative sample of individual-level studies on individual cognition in knowledge management. For instance, motivators such as job satisfaction and organizational commitment (Teh and Sun, 2012), increasing credibility (Wu and Zhu, 2012), success and promotion (Hendricks, 1999), reward system (Eze *et al.*, 2013), positive attitude (Roodbari *et al.*, 2016b), reputation and enthusiasm to help (Anitha, 2006) and learning and rewards (Yee, 2010; Jolaee *et al.*, 2014), each suggest potential motivating factors and support motivation for sharing as a key aspect of building knowledge repositories in an organization. These studies (alongside others) have also addressed the risks of knowledge sharing – negative motivators, albeit indirectly.

Beyond individual motivations to share knowledge, other studies recognize the importance of ability to share knowledge. For example, Eze *et al.* (2013) note the influence of confidence in interacting with colleagues, whereas Fathi *et al.* (2011) and Jolaee *et al.* (2014) mention self-efficacy as barriers to interaction.

Besides the motivations of knowledge sharing (benefits and risks) and ability to share knowledge, a particularly crucial factor in studies of individual cognition and knowledge management are situational constraints placed on individuals who wish to share knowledge. While constraints are barriers to interacting with others, they are distinct in that they do not directly refer to individual attributes and abilities such as self-efficacy. One previously suggested and particularly essential element, especially within the organizational context, is time constraint. Although prior research suggests that having the motivation and the requisite knowledge is necessary to support knowledge sharing (Trada and Goyal, 2020), numerous studies suggest that lack of time can inhibit knowledge sharing in organizations. Because exchanging, internalizing, combining or co-creating knowledge can be a demanding endeavor, individuals may understandably focus solely on work demands for which they have primary accountability (Connelly *et al.*, 2009).

These studies suggest that individual motivations alone are insufficient to influence knowledge sharing and as such provide support for the use of motivation–opportunity–ability as a three-dimensional framework for the CSE.

The common theme emerging from our understanding of the knowledge management literature is that three primary aspects exist (motivation, opportunity and ability), which

influence individual choices on whether to interact and exchange knowledge within organizations. Theoretical perspectives as well as empirical research suggest that motivation to share knowledge, opportunity to share knowledge and ability to share knowledge are key individual-level competencies that influence interaction between individuals (Connelly *et al.*, 2009). We introduce the CSE as the latent construct made up of the three dimensions of MOA.

2.3 Defining dimensions of capacity for social exchange: motivation/opportunity/ability

The MOA framework, originally introduced by MacInnis and Jaworski (1989) in the context of marketing communication effectiveness, has been found to be an important antecedent of knowledge sharing (Ardichvili, 2008; Siemsen *et al.*, 2008; Kettinger *et al.*, 2015; Radaelli *et al.*, 2014). In the marketing context, motivation implies audience interest and openness to advertising communication. Opportunity relates to the audience ability to focus and to allocate attention to the advertising vehicle (i.e. television, print ad, etc.). And finally, ability suggests skills, experience with the product or service and knowledge to understand and interpret the messaging (MacInnis *et al.*, 1991).

The MOA framework was extended to knowledge sharing research by Siemsen and Colleagues (2008). The authors established individual employee MOA to gauge the degree of knowledge sharing that can occur within an organization. Later, and relevant to our study, Turner and Pennington (2015) use the MOA framework as antecedent to organizational learning and ultimately to innovation success. Even more recently, Ahmad and colleagues support the MOA framework with social exchange theory and show that the three dimensions moderate the relationship between charismatic leadership and service-sales performance (Ahmad *et al.*, 2021).

2.3.1 Motivation. Although the specific elements that enhance motivation in the MOA framework vary across different studies, there is consistency in the importance of motivation in explaining knowledge sharing (Gruen *et al.*, 2005; Siemsen *et al.*, 2008). We leverage the definitions from the MOA framework's elements introduced by Siemsen *et al.* (2008) to define motivation as the primary driver of individual behavior, that is, the willingness to act. Motivation explains the successful practice of knowledge sharing among employees (Siemsen *et al.*, 2008). For instance, personal ambition (Hendriks, 1999), trust in organizations (Ardichvili, 2003), enjoyment (Hsu and Lin, 2008), rewards, self-efficacy (Lin, 2007), willingness (Lam and Lambermont-Ford, 2010) and altruism are elements of motivation that have been positively associated with knowledge sharing. Based on past studies, we assume motivation to be a key dimension that influences knowledge sharing.

2.3.2 Opportunity. Opportunity refers to constraints that affect an individual's ability to act, that is, to transfer knowledge. A common constraint in organizational culture is the availability of time to allow employees or groups to interact and create, or share, knowledge (Siemsen *et al.*, 2008). Other examples of situational impediments to opportunity discussed in the literature include fear of organizational criticism, fear of misleading organizations, extreme security environments (Ardichvili *et al.*, 2003), norm sharing (Gagné, 2009) and open egocentric networks (Reinholt *et al.*, 2011; AlHussan *et al.*, 2017; La Rocca and Snehota, 2014). Here, having the "right opportunity" is an important precondition for knowledge sharing. Following MacInnis *et al.* (1991) and Siemsen *et al.* (2008), we conceptualize opportunity as the availability of time to allow individuals to share knowledge. Opportunity (time) is considered one of the most crucial constraining factors in the knowledge management literature.

2.3.3 Ability. The third element of the MOA framework refers to the skill, proficiency and knowledge of the parties at hand in enabling effective exchange (MacInnis *et al.*, 1991). Ability represents an individual's skills or knowledge base in relation to action (Rothschild, 1999; Siemsen *et al.*, 2008). Having the ability or necessary knowledge is an underlying

assumption for sharing knowledge (Siemsen *et al.*, 2008). Beyond this, prior knowledge helps to build new knowledge, which can be absorbed in the workplace through experience, formal education and training, and is thus a prerequisite for interacting with colleagues (Reinholt *et al.*, 2011). Gruen *et al.* (2005) included past experiences, subjective knowledge, expertise and a sense of situational competence in their definition of ability (Gruen *et al.*, 2005). Human intellectual capital (ability) is recognized as an important dimension to successful innovation (Lee, 2001), and we follow prior research by defining ability as the “skills or knowledge related to the action” (Siemsen *et al.*, 2008, p. 427).

2.4 Mutual reinforcement of motivation, opportunity and ability elements: capacity for social exchange as a second-order latent concept

According to Siemsen *et al.* (2008) and Kettinger *et al.* (2015), motivation is considered to be the most vital of the three dimensions – although the remaining two elements (ability and opportunity/time) are by no means unimportant. For example, when motivation levels are high, ability has a positive effect; yet, if motivation levels are low, increasing ability has no effect (Clark *et al.*, 2005). Further, highly motivated groups can make poor decisions (Hackman, 1990; Janis and Mann, 1977), and the lack of opportunity can impede the productivity of even the most highly motivated employees (Rothschild, 1999). Turner and Pennington (2015) tested each dimension of MOA as antecedents to knowledge sharing in the context of corporate entrepreneurship and found motivation to be the strongest influencer and the other two dimensions to be highly correlated. The authors did not test the holistic association. Likewise, and more recently, Ahmad *et al.* (2021) studied the moderating effect of the three dimensions – MOA – between charismatic leadership and service to sales ambidexterity in a service to sales context. Motivation was found to have the strongest moderating influence, but all three dimensions were positive and significant. These scholars also did not test the holistic moderating influence of the MOA model.

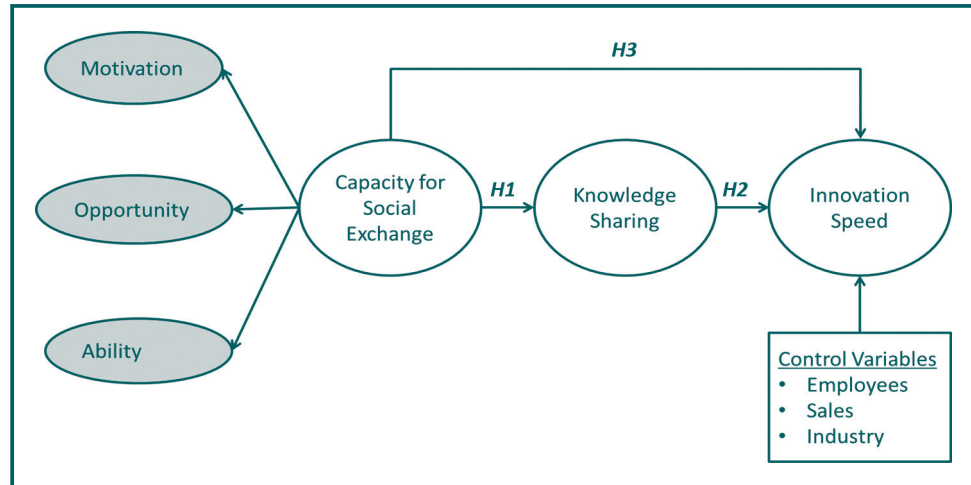
It follows that each of the three dimensions should be considered as important to the overall framework, as has been found in previous studies. In the present contribution, we go beyond the individual dimensions and refer to the mutually reinforcing elements of motivation, opportunity and ability as the CSE. As MacInnis *et al.* (1991) suggest, while motivation is the most significant factor in the framework, each of the elements is unique, important and interrelated within the MOA framework. Therefore, we posit that CSE is a second-order construct with three mutually complementary first-order dimensions: motivation, opportunity and ability. Prior studies using the MOA framework fail to fully investigate the holistic influence of the three dimensions in addition to the importance of each dimension on its own.

3. Hypothesis development

Figure 2 presents the hypothesized research model. We discuss the hypotheses below.

3.1 Capacity for social exchange and knowledge sharing

Knowledge sharing refers to the activities of transferring knowledge between individuals or groups (Lee, 2001). Chiu *et al.* (2006) found that a lack of social capital could negatively affect knowledge sharing in virtual communities, which bears important implications for this study because the supply chain within a service industry operates both face-to-face and virtually. We follow the definition of knowledge sharing introduced by Siemsen *et al.* (2008) in that knowledge sharing involves at least two people: a sender (here, an employee who attempts to share knowledge) and a recipient (the coworker who is intended to acquire it). In this context, we seek to understand and focus on what motivates a sender to transfer expertise to others.

Figure 2 Research model

The core emphasis in social exchange theory lies in individual cognition, which includes the perceived benefits and risks of interacting with others and sharing knowledge. The former of these are the subjective gains that accrue to an individual from interacting with others, whereas perceived risks are the subjective losses incurred from sharing knowledge. According to [Blau \(1964\)](#), individuals are more willing to share expertise when they believe that it will not distract them from their primary work and will result in tangible rewards (bonuses) or intangible rewards (status, enjoyment and promotions), over time. [Davenport and Prusak \(1998\)](#) argue that motivation to share knowledge is a result of perceived benefits, and [Wasko and Faraj \(2005\)](#) go on to note that the possibility of improving reputation (a perceived benefit) motivates advice-giving and affects the frequency of interaction.

Thus, CSE – the measure of the degree to which individuals are motivated, capable and have the opportunity to interact – should positively relate to knowledge sharing. Having the necessary motivation suggests that organizational mechanisms have been put in place to encourage knowledge sharing. For example, Siemens uses a points-based rewards system to recognize individuals who meaningfully contribute to eliminating re-work. The provision of adequate time – hence, opportunity – in which to share knowledge is also suggestive of organizational priorities.

Once provided with the appropriate incentives and opportunity, capable individuals might perceive the personal benefits in sharing as outweighing its risks. Additionally, depending upon the strategies used to augment opportunities (such as flexible work arrangements), individuals may also perceive management as genuinely acting in the best interest of employees, thereby identifying or bonding with the organization and reciprocating goodwill by sharing expertise to the benefit of all. It stands to reason that if the separate dimensions of MOA have been found to be positively associated with knowledge sharing, then the holistic construct of CSE should be strongly associated with knowledge sharing. Therefore, we propose the following hypothesis:

H1. CSE is positively related to knowledge sharing.

3.2 Knowledge sharing and innovation speed

Innovation speed is defined as the period of time between the launch of a new idea and the placement of a product (or service) in a market ([Kessler and Chakrabarti, 1996](#)). According

to a systemic review of knowledge sharing and innovation, the positive relationship between the two variables has been frequently studied, as early as 1973 (Castaneda and Cuellar, 2020). Knowledge sharing has been linked to innovation and innovation speed and, ultimately, to positive firm performance (Spencer, 2003; Wang and Wang, 2012; Chen and Huang, 2009; Lin, 2007; Inemek and Matthyssens, 2013). Wang and Wang (2012) find that the speed of innovation at successful B2B firms derives from both explicit and tacit knowledge. Resource and knowledge collaborations have been found to improve the capacity for introducing a new product (Zhao *et al.*, 2020; Cao and Zhang, 2011), and acquiring knowledge from consultants as knowledge brokers was found to produce innovation at a high rate of speed (Hargadon, 1998). In other words, the quality of information sharing (IS; knowledge sharing) in a positive relationship with partners results in more rapid innovation.

In other contexts, knowledge sharing between individual team members has been found to be positively related to team-level innovation (Hu and Randel, 2014). Previous studies have shown that innovation can be improved when an organization encourages individuals to share knowledge freely (Spencer, 2003). Innovation performance and knowledge sharing have a positive relationship, especially when gaining external knowledge from partners is crucial, for example in supply chains (Ritala *et al.*, 2014), where collaborative innovation activities are also positively affected by knowledge sharing (Wang and Hu, 2020). Thus, given the overwhelming evidence, we propose the following hypothesis:

H2. Knowledge sharing is positively related to innovation speed.

3.3 Capacity for social exchange and innovation speed

Based on prior research, we would expect CSE to improve personal innovativeness (e.g. the number of innovative ideas proposed by individuals); however, our context suggests that CSE may not enhance innovation speed at the organizational level. Specifically, we are interested in innovations that are implemented with the purpose of achieving organizational objectives and improving work processes and services. Thus, the emphasis here lies on innovations that affect not only the individual but also groups that make use of services and work processes. As Van de Ven (1986) suggests organizational innovations do not take place in isolation: executives, close working colleagues and other important stakeholders such as strategic customers are affected by innovation and attempt to steer novel ideas in a direction that is acceptable to everyone. Similarly, Scott and Bruce (1994) have stressed the importance of interacting with colleagues and eliciting support for new ideas. Amabile (1988) proposed a model of creativity that highlights the importance of a work environment that includes top management support and the discussion of ideas among colleagues. This implies that while motivation, opportunity and ability may indeed result in the development of novel ideas, in organizational contexts, these three dimensions alone do not suffice to convert ideas into practice. Therefore, unless discourse on new ideas and frequent interactions take place, generating support from other significant stakeholders becomes difficult and may restrain or slow the implementation of innovative ideas. In an organization, innovation does not happen without the collaboration of multiple organizational units and individuals.

Therefore, we believe that in the absence of knowledge sharing, any benefits of CSE are likely to emerge in form of new ideas and thoughts that do not translate into value creation for the company. In the absence of interactions with others, motivation, opportunity and ability are, by themselves, not strong enough to generate support – without which the perception can be created that an idea generator is “going against the crowd,” thereby risking sanctions should the proposed idea be deemed inadequate. Furthermore, when others, especially higher executives, show indifference to fresh ideas, the benefits of

proposing novelty do not outweigh the immediate personal risks of proposing them. Accordingly, we believe in the following hypothesis:

H3. Knowledge sharing entirely mediates the relationship between CSE and innovation speed.

3.4 Knowledge channels as boundary objects to facilitate knowledge sharing

In this study, we also consider KC as an organizational support mechanism and explore their influence on both sharing as well as speed of innovation.

KC refer to the boundary-spanning strategies used by organizations that permit the transfer, recombination and/or creation of knowledge among supply chain partners (Rosenzweig and Roth, 2007). According to Rosenzweig and Roth, who outline the concept, “KC enables the cross organizational integration required for creating formal and informal connections and also the potential for deep integration among partners” (p. 1317). KC therefore may be thought of as a coordinating or integrating mechanism used by partner organizations to bring together the various pockets of knowledge that reside within a supply chain’s diverse functional units and to leverage shared expertise and interactions so as to identify opportunities and innovate across products, services or processes which help them stay competitive (Uzzi and Lancaster, 2003). KC may be formal (e.g. the deployment of electronic interfaces) or informal (e.g. periodic informal gatherings or cross-organizational points of contact, where functional teams from different partners may be co-located to facilitate exchange and the sharing of expertise). Regardless of the specific form of deployed KC, the literature generally concurs that KC allow knowledge to converge, thereby providing a forum that enables the clarification, debate and enactment of knowledge among partners.

KC are important for the sharing of information because boundaries between organizations can adversely impact information flow and knowledge. For instance, the extent of information shared between partners can be influenced by the extent to which parties mutually depend upon each other for their business needs. Furthermore, boundaries may act as barriers to that helps organizations stay competitive. Beyond economic considerations, the literature suggests that people belonging to different organizations possess different knowledge sets and, often, non-overlapping mental schemas (Carlile, 2002, 2004), which can impede the exchange of information between partners due to limited absorptive capacity (Cohen and Levinthal, 1990). Thus, integrating and coordinating mechanisms in the form of KCs are crucial to enabling members of different organizations with unique mental schemas to understand, interpret and make sense of novel information (Claro and Claro, 2010). Indeed, Carlile (2004) argues that KC are critical to sustaining the intense sharing of knowledge because the very act of constructing these boundary-spanning channels requires the creation of a shared commitment and the establishment of routines, stories, models or a common language that allows members from distinct organizations to contribute to shared objectives (Kellogg et al., 2006).

In the context of this study, KC capture the informal mechanisms that an organization’s individual managers may use to share expertise. In the governance branch of organization theory, these informal practices of co-locating team members or participating in frequent informal gatherings where knowledge can be shared are often labeled as “clan controls” (Ouchi, 1979). Knowledge sharing has been found to enable sharing and increase innovative activity because they allow organizations to overcome the three types of boundaries outlined by Carlile (2004) that prevent the exchange and coordination of dispersed knowledge in supply chains. Shared mental schemas that arise from frequent interactions may overcome syntactic boundaries (differences in information and knowledge content) and semantic boundaries (differences in viewpoints and priorities). Finally, time spent together engaging with partners also builds quasi-clan types of working relations not exclusively dominated by economic interests but which also emphasize cooperative, mutually beneficial goals important for overcoming pragmatic boundaries (Homans, 1958;

Emerson, 1976). Therefore, we expect the presence of KC to enhance knowledge sharing by generating a greater breadth of individuals amongst whom knowledge may be exchanged, as well as more frequent opportunities in which this can occur. Accordingly, we hypothesize as follows:

H4. KC positively moderate the relationship between CSE and knowledge sharing.

3.5 Knowledge channels as boundary objects to facilitate innovation speed

We expect KC to moderate the relationship between CSE and innovation speed for several reasons. The literature agrees that communication between individuals who possess diverse opinions, viewpoints, and information is a key factor driving innovation. Although prior work suggests that communication among individuals with diverse viewpoints and knowledge can be a source of conflict and may be counterproductive, *yet also* that the conflict process may prove to be fruitful in facilitating innovation. Conflict can bring hidden assumptions and agendas to the fore (Flaatin, 2007), and productive debate may result in a broader discovery of ideas and resolutions to potential problems and disagreements (Tjosvold *et al.*, 2014). As argued above, by instituting informal meetings and co-locating partners, KC open opportunities for individuals with diverse pockets of knowledge and mental schemas to come together and collaborate. Although individual creativity and talent are certainly important, by themselves these remain insufficient as conditions for innovation for they do not present fruitful opportunities for collaboration. KC augment creativity and talent by providing realistic opportunities for knowledge sharing and innovation (Del Giudice *et al.*, 2014, 2015; Scuotto *et al.*, 2017). Moreover, in the context of knowledge sharing among supply chain partners, KC may be crucial for organizing both financial and non-financial resources during the implementation phase. For instance, KC may serve to provide social capital resources (adoption of ideas by stakeholders) that increase the likelihood of gaining access to additional material resources. Therefore, we hypothesize as follows:

H5. KC positively moderate the relationship between CSE and innovation speed.

4. Methodology

4.1 Participants and procedure

The sampling frame of this research consisted of service businesses in the USA, and the data were collected through Zoomerang, an online survey research firm; the B2B panel of Zoomerang formed the sampling frame. Other studies have successfully used data from Zoomerang and suggest that the data provided there are valid for research purposes (Iskandar *et al.*, 2015; Pham *et al.*, 2015; Trzesniewski and Donnellan, 2014). The respondents were screened to ensure a sampling of only full-time and mid-to-high-level managers and executives in the service industry. A total of 1,500 respondents were contacted, and 264 complete responses were received. The response rate of 17.6% is similar to other B2B studies using the Zoomerang panel (Glenn Richey *et al.*, 2010). The sample included 97 (37%) public and 167 (63%) private firms. Service industries included were 70 (27%) health-care and social work industries; 50 (19%) finance, banking and insurance; and 45 (17%) retail trade; remaining responses stemmed from the information, transportation and telecommunication industries. The respondents' job titles encompassed 144 (53%) managers and assistant managers, and 179 (53%) respondents worked for small- and medium-sized companies that employed fewer than 500 individuals (Allocca and Kessler, 2006). In addition, 217 (82%) respondents worked at organizations with total sales of less than US\$100m. No significant statistical differences were observed regarding job levels, total sales, number of employees or industries. No non-response bias was found when comparing early and late respondents in the targeted demographics. Furthermore, there were no missing data, as we used forced responses in a Web-based survey.

4.2 Measures

All the constructs in our research model (Figure 1) have reflective items, where the causal arrow points from the latent construct to the observed variables (Jarvis *et al.*, 2003). The constructs measured in this study include CSE, knowledge sharing, KC and innovation speed. The operationalization is provided in Table 1. The CSE construct was conceptualized as a second-order construct with three first-order dimensions: motivation, opportunity and ability, the items for which were adopted from Siemsen *et al.* (2008). These items were measured on a five-point Likert scale (1 = “Strongly disagree,” 5 = “Strongly agree”) and captured employees’ willingness, time availability and proficiency to share knowledge within a supply chain.

Innovation speed measurement items were modified from Nassimbeni (2003) to fit the study at hand. The innovation speed items captured the ability of a firm to develop new services – or add new features to existing services – at a higher rate than its competitor. The items of innovation speed used a five-point Likert scale to assess a firm’s position relative to its competitors (1 = “Far behind,” 5 = “Far ahead”).

The knowledge channel items were adapted from Rosenzweig and Roth (2007) and captured the boundary-spanning activities used in the supply chain to create, recombine and transfer knowledge. The items for KC used a five-point Likert scale (1 = “Strongly disagree,” 5 = “Strongly agree”).

The knowledge sharing scale was adapted from Siemsen *et al.* (2008) and captured the efforts and attempts made by supply chain partners to share knowledge. The knowledge sharing construct is measured on a five-point Likert scale (1 = “Strongly disagree,” 5 = “Strongly agree”). See item/scale detail in Appendix.

4.3 Control variables

We included three control variables in our research model: number of employees, sales and industry. Number of employees is a measure of firm size and allows partially canceling out the effect of economies and dis-economies of scale. Sales as a control allows controlling for differences in resources among firms. Finally, industry as a control variable allows controlling for difference in innovation levels across different industries. Some combination of these three control variables is commonly used in related research (Johnson and Sohi, 2017; Wang and Wang, 2012; Zimuto *et al.*, 2018; Kownatzki *et al.*, 2013).

Table 1 Operational definitions

Construct name	Operational definition	Source
Capacity for social exchange	Organizational culture and structure to advocate workplaces for sharing knowledge	Majiros (2013)
Motivation	Driver of knowledge-sharing behavior and willingness to share knowledge	Siemsen <i>et al.</i> (2008)
Opportunity	Availability of time to share knowledge	Siemsen <i>et al.</i> (2008)
Ability	Skills or proficiencies related to sharing knowledge	Siemsen <i>et al.</i> (2008)
Knowledge sharing	Acts and attempts to send knowledge	Siemsen <i>et al.</i> (2008)
Knowledge channels	Boundary-spanning strategies that allow the transfer, recombination, or creation of knowledge	Rosenzweig and Roth (2007)
Innovation speed	Development of innovative products and services at a faster pace than competitors	Nassimbeni (2003)

5. Results

5.1 Convergent and discriminant validity

We assessed the convergent and discriminant validity of the measurement instrument by using the procedure suggested by Hair *et al.* (2016) (see Table 2). The average variance explained (AVE) for each of the constructs in our research model was above 0.5, thus indicating convergent validity. Moreover, the square root of the AVE was larger than all inter-construct correlations for all the scales, thus indicating discriminant validity. CFA analysis was also used to assess the validity of the constructs. Item loadings for all the constructs were highly significant, thus providing further evidence of convergent validity.

5.2 Hypotheses testing

The hypotheses test was performed through SEM by using AMOS 21 software. The overall fit of the structural model was quite satisfactory (χ^2 (df) = 247 (135), $p \leq 0.001$; CFI = 0.976; RMSEA = 0.056; SRMR = 0.058), indicating the nomological validity of our hypothesized model.

Our *implicit hypothesis* that CSE is a second-order construct with three first-order dimensions – motivation, opportunity and ability – was supported. The overall fit indices of the hypothesized second-order factor structure were excellent (χ^2 (df) = 57 (24), $p \leq 0.001$; CFI = 0.986; RMSEA = 0.073; SRMR = 0.034). The path loadings of the three first-order factors – motivation, opportunity and ability – were 0.963 ($p \leq 0.001$), 0.251 ($p \leq 0.001$) and 0.840 ($p \leq 0.001$), respectively.

The direct effect hypotheses were evaluated by using the path loadings obtained through the SEM analysis. *H1*, which states that CSE is positively related to knowledge sharing, was supported ($\beta = -0.144$, $p \leq 0.308$). *H2*, which states that knowledge sharing is positively related to innovation speed, was supported ($\beta = 0.673$, $p = 0.008$). *H3*, which states that CSE is positively related to innovation speed, was not supported ($\beta = -0.180$, $p \leq 0.379$).

While *H3* was not supported, the first-order correlation between CSE and innovation speed was highly significant ($r = 0.407$, $p \leq 0.001$). This indicates a net suppression effect, where the large correlation between CSE and knowledge sharing ($r = 0.942$, $p \leq 0.001$) cannibalizes the relationship between CSE and innovation speed, thus making it insignificant.

We also evaluated the impact of three control variables (number of employees, sales and industry) on the research model. The effects for all three control variables were not significant. Also evaluated was the effect of an *a priori* marker variable borrowed from leisure literature on the research model. The inclusion of the marker variable had no significant effect on the results.

Table 2 Descriptive statistics: composite reliabilities (CR), AVE and correlations ($n = 264$)

Variables	CR	AVE	1	2	3	4	5	6	7
1. Industry	–	–	–						
2. Employees	–	–	–	–					
3. Sales	–	–	–	–	–				
4. Motivation	0.960	0.889	0.024	0.081	0.017				
5. Opportunity	0.886	0.723	0.083	0.126	0.155	0.194			
6. Ability	0.958	0.883	–0.024	0.052	0.018	0.811	0.305		
7. Knowledge sharing	0.958	0.884	0.037	0.087	0.017	0.908	0.269	0.784	
8. Innovation speed	0.938	0.835	0.093	0.074	0.031	0.378	0.290	0.374	0.460

5.3 Test of moderation effects using process analysis

We evaluated the moderation effects by using the process Model 8 macro suggested by Hayes (2017), which is used to test models with moderated mediation effects. Figure 4 provides the complete moderated mediation model, where CSE is the independent variable, information sharing (IS) is the mediator, innovation speed (IN) is the dependent variable and knowledge channels (KC) is the moderating variable. The statistical model represents two linear equations for the outcome variables – information sharing and innovation speed – as provided in equations (1) and (2):

$$IS = i_{IS} + a_1(CSE) + a_2(KC) + a_3(CSE * KC) + a_4(Employees) + a_5(Sales) + a_6(Industry) + e_{IS} \quad (1)$$

$$IN = i_{IN} + c'_1(CSE) + c'_2(KC) + c'_3(CSE * KC) + b_1(IS) + b_2(Employees) + b_3(Sales) + b_4(Industry) + e_{IN} \quad (2)$$

Table 3 provides the results of the PROCESS analysis. *H4*, which states that KC positively moderate the relationship between CSE and knowledge sharing, was not supported ($\beta = -0.001$, $p = 0.976$). On the other hand, *H5*, which states that KCs positively moderate the relationship between CSE and innovation speed, was supported ($\beta = 0.1108$, $p = 0.013$). In addition, the index of moderated mediation (a_3b_1) was not significant, as ($a_3b_1 = 0.1487$, 95%CI = 0.0312 to 0.2663) 95% bootstrap confidence interval does not include zero. Therefore, we only have evidence of moderation of the direct effect of CSE on innovation speed, whereas the moderated mediation effect of CSE on innovation is not significant.

Simple slopes were used to evaluate the conditional direct effect of CSE on innovation speed, based on the levels of KC. Figure 3 provides the simple slope representation of the conditional effect. The simple slope of CSE with innovation speed at a low level of KC was not significant ($\beta_{low} = 0.0443$, $p = .5991$). The simple slope of CSE with innovation speed at a medium level of KC was barely significant ($\beta_{medium} = 0.1441$, $p = 0.0508$). The simple slope of the relationship of CSE with innovation speed at a high level of KC was highly significant ($\beta_{high} = 0.2439$, $p = 0.0035$). It is evident from the figure that the effect of CSE on innovation speed increases with the level of KC (Figure 5).

Table 3 OLS regression coefficient with confidence intervals (standard errors in parentheses) estimating information sharing and innovation speed

Variables	Information sharing (IS)			Innovation speed (IN)		
		Coeff	95% CI		Coeff	95% CI
Capacity for social exchange (CSE)	$a_{1\rightarrow}$	0.8920*** (0.0000)	0.5772, 1.2069	$c'_{1\rightarrow}$	-0.2026 (0.2001)	-0.5131, 0.1080
Information sharing (IS)				$b_{1\rightarrow}$	0.1248* (0.0326)	0.0104, 0.2392
Knowledge channels (KC)	$a_{2\rightarrow}$	0.3363 (0.1574)	-0.1307, 0.8032	$c'_{2\rightarrow}$	-0.3894 (0.0803)	-0.8261, 0.0473
CSE × KC	$a_{3\rightarrow}$	-0.0014 (0.9763)	-0.0946, 0.0918	$c'_{3\rightarrow}$	0.1108* (0.0126)	0.0239, 0.1976
Company size (U_1)	$a_{4\rightarrow}$	0.0100 (0.6253)	-0.0304, 0.0504	$b_{2\rightarrow}$	0.0001 (0.9975)	-0.0376, 0.0377
Sales (U_2)	$a_{5\rightarrow}$	-0.0692 (0.1542)	-0.1645, 0.0261	$b_{3\rightarrow}$	-0.0152 (0.7369)	-0.1044, 0.0739
Industry (U_3)	$a_{6\rightarrow}$	-0.0049 (0.7796)	-0.0397, 0.0298	$b_{4\rightarrow}$	0.0108 (0.5122)	-0.0216, 0.0432
Constant	$i_M \rightarrow$	-0.4539 (0.5618)	-1.9929, 1.0850	$i_Y \rightarrow$	3.3247*** (0.0000)	1.8902, 4.7592
		$R^2 = 0.6697$			$R^2 = 0.2557$	
		$F(6,257) = 86.8466$, $p < 0.001$			$F(7,256) = 12.5644$, $p < 0.001$	

Notes: + $p < 0.01$; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$; OLS = ordinary least squares

Figure 3 Research model with path loadings

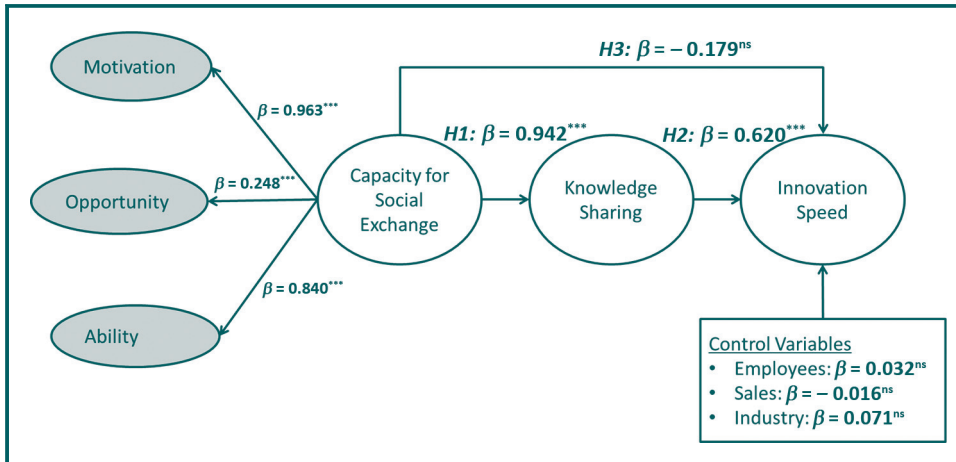
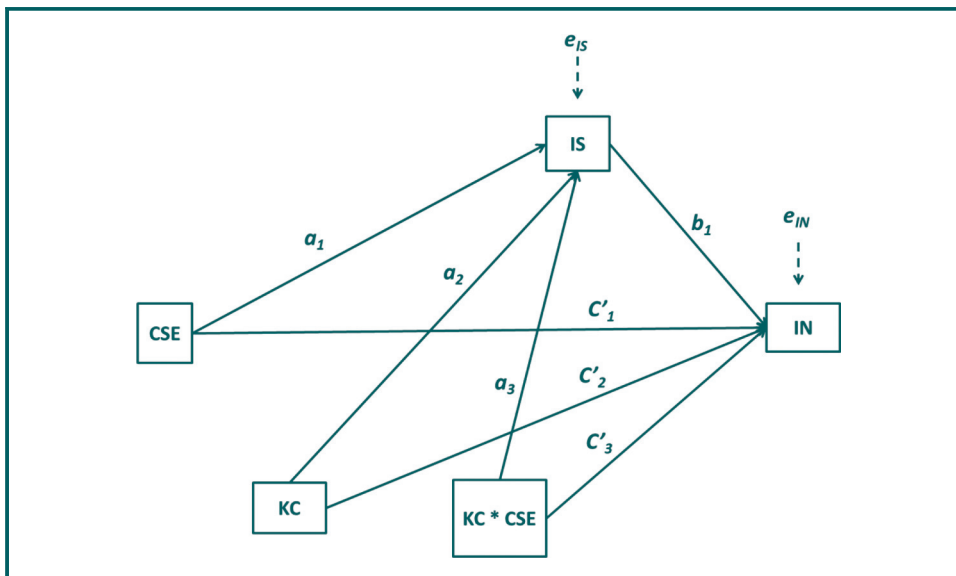


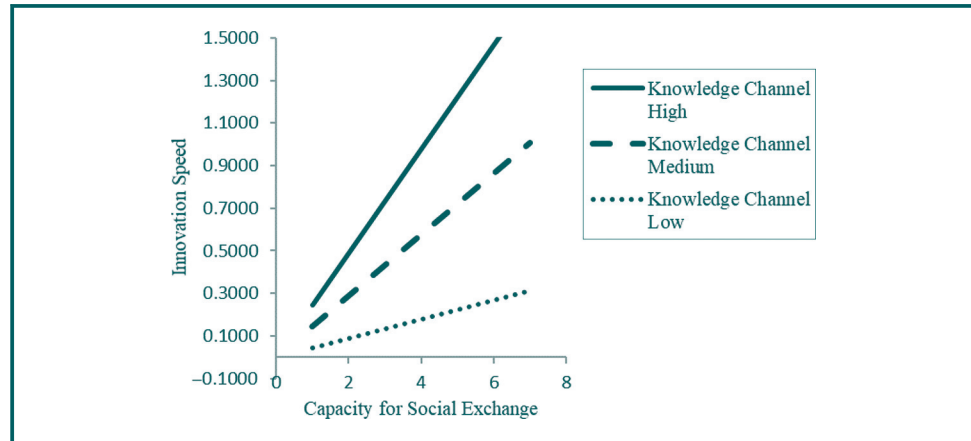
Figure 4 Moderated mediation model



6. Discussion and implications

We find that CSE is positively related to knowledge-sharing behavior ($H1$ is supported), hence suggesting that when organizations provide the right conditions (additional time or reduced workload), individuals may perceive that an organization values the exchange of knowledge. In particular, motivation (0.963) and ability (0.840) appear to be critical to knowledge sharing. Second, we find that knowledge sharing is positively related to sustaining innovation speed in service contexts ($H2$ is supported). For instance, through its consulting business, International Business Machine Corporation builds better relationships with its customers and improves their ability to innovate (Papailiou *et al.*, 2007). Third, as expected, we find an insignificant relationship between the CSE and innovation speed, thereby implying that knowledge sharing entirely mediates this relationship. It appears that capable and motivated individuals may fall short in either producing novel ideas or selling

Figure 5 Simple slope analysis



them to other stakeholders, opportunity notwithstanding – perhaps due to limits on individual rationality, efforts to develop ideas from inception to completion remain inadequate. Therefore, sufficient innovation speed is sustained when an employee's degree of sharing knowledge is increased through the increased CSE.

We also examined how context affects innovation speed through its interplay with the individual-level factors included in this study. Our hypothesis that KC positively moderates the relationship between CSE and knowledge sharing was not supported, possibly because the presence of CSE already establishes organizational support for activities such as knowledge sharing. For example, where CSE is high, organizations may provide reduced workload, additional “innovation and thought time,” or alternate methods of work (work from home and flexible work arrangements) to provide time to engage in sharing behaviors. Finally, our hypothesis that KC positively moderates the relationship between CSE and innovation speed was supported. Thus, we found that, in the absence of KC, CSE has no significant effect on innovation speed; for high levels of KC, however, an increase in CSE can result in increased speed of innovation.

Our result for the interaction between KC and CSE can be interpreted in terms of limits on rationality, which is consistent with the hypothesis that CSE increases the sharing of knowledge. Innovation usually takes place in two stages, and the mere generation of a novel and promising idea does not ensure its implementation and value generation. Even highly motivated and capable individuals are subject to their own bounded rationality – without the input of colleagues, an individual may not independently be able to sense opportunities to innovate. Furthermore, and due to limited rationality, individual employees may possibly perceive their own ideas as lacking value, or they may fear failure and resist aggressively pushing their ideas up the ranks or horizontally to partners within the supply chain. Accordingly, it makes sense that CSE increases knowledge sharing yet may not necessarily increase the rate of innovation. For innovation speed to increase, organizations must institutionalize strategies such as KC for boundary spanning and actively communicate their vital role in remaining competitive.

6.1 Theoretical implications

This study provides a number of theoretical contributions. Recently, concerns over the relative lack of interest in understanding knowledge sharing from a micro-level perspective have been raised in both the knowledge management literature (Felin and Foss, 2005; Amabile, 1996) and B2B literature (Alinaghian *et al.*, 2020; Yang *et al.*, 2019). We thus

contribute by answering calls (Forkmann *et al.*, 2018; Alghababsheh and Gallea, 2020) to conduct more studies that examine knowledge management at the individual level, as well as on the interaction of individuals and context. Based on social exchange theory, our findings lend support to the buyer–supplier relationships from a micro-level perspective and reveal that external relationships with suppliers present a potential starting point and play a catalytic role in developing innovations more rapidly. Our study is an initial attempt to identify the micro-level antecedents of innovation speed and to examine these empirically.

Second, we introduce a second-order latent construct – the CSE – to the extant literature. Weick *et al.* (2005) propose that a broad multidimensional construct with specific dimensions such as CSE can be used as a platform for further theory building, as well as for making sense of some of the perplexing and contradictory findings in prior research. Past research has clearly established the role of knowledge sharing in buyer–supplier relationships on innovation performance, with varying success. Some studies find that supplier integration helps firms accelerate the development process of new products, enhance their quality and reduce the costs of innovation (Lau *et al.*, 2010). Other studies find that supplier integration may lead to negative consequences (e.g. coordination problems) and jeopardize firms' valuable knowledge (Wagner and Hoegl, 2006). Our CSE construct sheds light on these contradictory findings that suffuse the extant literature.

Third, by introducing the idea of CSE and applying the MOA framework, we contribute by exploring how individual-level variables relate to the sharing of knowledge and the organizational competencies that encourage and facilitate this exchange of knowledge, which, in turn, affect a firm's speed of innovation. Prior research has not investigated how employees' motivation, opportunity and ability to share knowledge relate to knowledge sharing attempts and innovation speed. Our findings suggest that information exchange in buyer–supplier relationships is not resolved only by the readiness but also the capacity, for social exchange. Past research has argued that tacit knowledge sharing could be difficult without the assistance of clarification and description (Hansen, 1999). A strong CSE in buyer–supplier relationships thus plays a role in helping the receiving party to understand and use tacit knowledge better, thereby simplifying the relationship and, hence, smoothing the transfer of tacit knowledge that leads to faster innovation (Reagans and McEvily, 2003; Rowley *et al.*, 2000).

Finally, this study incorporates KC as an informal mechanism and investigates how it interacts with CSE in terms of knowledge acquisition and innovation speed. Commonly, manufacturers attempt to cut costs and enhance delivery performance by working with suppliers (Hoegl and Wagner, 2005). However, suppliers are increasingly regarded as important sources of information during the innovation process (Fliess and Becker, 2006), therefore emphasizing the importance in this process of identifying the conditions under which suppliers are most willing to exchange information with manufacturers.

6.2 Managerial implications

B2B firms should engage in knowledge sharing with their suppliers during the innovation process. This study offers valuable implications for managers on how to collaborate with their suppliers and develop effective communication and knowledge-sharing practices that help firms develop innovations more rapidly. Our results suggest that, prior to launching knowledge-sharing practices, managers should ensure the presence of motivation, opportunity and ability among employees involved in the knowledge-sharing process. Without the existence of these three elements of CSE, bilateral routines of knowledge sharing can hardly be initiated. Strong CSE in buyer–supplier relationships helps buyers to obtain and use fresh information and generates opportunities for them to improve their products or services faster than those with weak CSE. With a high level of CSE, buying firms are more capable of learning and experimenting with various alternatives, which can enhance their ability to adapt effectively and promptly to market changes. Moreover, our

results highlight the need for the effective use of KC for information acquisition and dissemination. Managers should focus on how to overcome cultural and organizational barriers that obstruct KC and how to integrate new communication tools into existing channels.

7. Limitations and future research

The interpretation of our findings should be considered alongside our study's limitations. The research at hand employed a subjective self-reported measure for innovation speed – although we sampled knowledgeable executives, we cannot rule out the possibility of bias. Future studies may wish to include objective measures for innovation speed, for example by examining patent databases. A second limitation of this study is that we left aside aspects of knowledge sharing that could conceivably contribute to rapid innovation; for example, according to [Kessler and Bierly \(2002\)](#), quality of ideas is an important antecedent for organizational innovation speed. Beyond this, our measure of CSE could be enriched by including other kinds of opportunities. Due to considerations of the limited space available here, we measured the availability of time. However, material and social capital resources may be included as additional measures to enhance this construct in further research. Knowledge management and organizational performance are rewarding and significant topics for study, for motivating employees to share knowledge is essential if organizations want to innovate and remain competitive. Our study proffers a small yet vital step in this direction. Further, in the study we followed prior organizational research in selecting our controls. Future research could consider controls at the individual as well as organizational levels.

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Further reading

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Appendix. Scale descriptions and reliability statistics

Siemens *et al.* (2008), Nassimbeni (2003), Rosenzweig and Roth (2007)

Motivation to share

Please respond to Items 1–4 about the motivation level of you and your supply chain partners to share knowledge.

1. We have no intention to share our knowledge with our supply chain partners.
2. We are motivated to share what we know with our supply chain partners.
3. We really want to share our knowledge with our supply chain partners.
4. We mean to share our knowledge with our supply chain partners.

Time availability

Please respond to Items 5–7 about the amount of time available to share knowledge in your business unit.

5. We have little free time to allocate during work.
6. We are usually under high time pressure at work.
7. The extra time we have available at work is limited.

Ability to share

Please respond to Items 8–10 about your business unit's ability to share knowledge with your supply chain partners.

8. We have the ability to transfer our knowledge to our supply chain partners.
9. We have the means to share our knowledge with our supply chain partners.
10. We are capable of sharing our knowledge with our supply chain partners.

Knowledge sharing attempt

Please respond to Items 11–13 about the attempts made by you and your supply chain partners to share knowledge.

11. We attempt to teach our knowledge to our supply chain partners.
12. We make an effort to transfer our knowledge to our supply chain partners.
13. We try to share our knowledge with our supply chain partners.

Knowledge channels

Please respond to Items 14–16 related to knowledge management practices of your business unit.

14. Our employees regularly visit supply chain partners to enable two-way sharing of expertise.
15. We are able to capture best practices via an information source that spans company boundaries.
16. We are able to leverage after-sales service activities as a source of cross-company-learning about customers.

Innovation speed

How do you compare with your competitors on the below-mentioned capabilities related to innovation? (Items 17–19)

17. How does your business unit's ability to develop new services at a high rate compare with your competitors?

18. How does your business unit's ability to develop new features in your existing services at a high rate compare with your competitors?

19. How does your business unit's ability to develop new service delivery technology at a high rate compare with your competitors?

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